SANJAYA

Remaining work to cover the SIH problem statement, expected solution and USPs

Submission readiness register | 28 September 2026 | Based on PRD.md and PROTOTYPE_AUDIT.md

Honest current position. The engine demonstrates blind BPSK/QPSK recipe recovery, a layered proof and certificate, generated data-sufficiency and stitching examples, and a gated receiver export. It does not yet demonstrate the full automated FSK/QAM/PSK + four-interleaver + four-FEC chain on independent real captures. Passing unit tests establishes implementation behavior, not the PRD accuracy, speed or false-certificate targets.

1. Read this before making a submission claim

• Working: five curated synthetic single-recording samples complete the blind proof path; synthetic stitching passes; a signed certificate can be rechecked when paired with its source recording. The most recent full engine check passed 108 tests. The UI build passed; those checks are not a browser usability study.

• Partial: six public HF recordings reach family/rate identification but no proven payload. Known-recipe demodulation and decoding cover more families than the blind search. Standalone FSK and convolutional interleaver tests have not been connected to the end-to-end recipe path.

• Unvalidated: numerical PRD KPIs, empirical false-certificate bound, real multi-site stitching, executed GNU Radio flowgraph, 8 GB streaming, 100 MB/2 s GUI response and offline installers.

• Claim rule: distinguish an implemented operator, a blind identification result, a known-recipe result, a real-file demonstration and a measured KPI. An undecided result needs its observed evidence and an accurate reason. Never infer missing data length from a frame that was not found.

2. Literal problem statement and expected solution: coverage map

Input: .IQ and .wav [PARTIAL]

Today: Common mono/stereo WAV, selected raw IQ and SigMF can be loaded; 100 MiB API cap.

Remaining: Full WAV and raw-IQ variant matrix, confident dtype/endian/IQ-order sniffing, RF64/BW64, metadata, 8 GB streaming and safe errors.

Done when: 200-file format bench and fuzz gate pass; all ambiguous formats show an override. See R-01.

(i) Sampling frequency and signal characteristics [PARTIAL]

Today: Declared sample rate, symbol rate, band, CFO, clock ppm, SNR and some FSK tone plans are measured.

Remaining: Headerless sample-rate backsolve; per-family estimators; uncertainty intervals and independent accuracy/coverage checks.

Done when: R-02/R-03 measurement targets pass on a held-out corpus, or the UI explicitly says unknown.

(i) Modulation, FEC and interleaving identification [PARTIAL]

Today: Blind BPSK/QPSK, convolutional K=3–9 and block geometry on a generated frame family; broader known-recipe operators.

Remaining: Blind 8PSK/QAM/OQPSK/MSK/FSK; RS/LDPC/concatenated/punctured FEC; convolutional/diagonal/pseudo-random interleaver search; calibrated abstention.

Done when: R-04/R-05/R-06 targets pass on sealed synthetic and independent real files.

(ii) Demodulate FSK, QAM and PSK [PARTIAL]

Today: Known-recipe PSK/QAM soft-demod; standalone 2/4/8-FSK tone-bank fixtures.

Remaining: Wire FSK into recipes, search, proof and API; add OQPSK/MSK, HF equalization, burst acquisition and oracle BER curves.

Done when: Each advertised family produces a verified bitstream through the GUI and API; R-07 curve and burst targets pass.

(iii) De-interleave four specified types [PARTIAL]

Today: Block, diagonal and pseudo-random inverse with supplied recipes; standalone finite-stream convolutional inverse.

Remaining: Connect convolutional interleaving to the end-to-end chain; blind identification for all four; manual override.

Done when: Bit-exact round trips and at least one full decode/proof per type; R-08/R-06 gates.

(iv) FEC: Viterbi, RS, concatenated, LDPC [PARTIAL]

Today: All four known-recipe decoders have tests; RS, concatenated and LDPC feed proof.

Remaining: Blind selection, variant coverage, puncturing/RS basis/LDPC catalogue; independent FER oracle curves and production kernels.

Done when: R-09 FER gap and R-05 blind-ID targets pass, with measured envelope stated.

(v) Bit-stream correlation; header and payload [PARTIAL]

Today: Blind sync/frame/CRC/counter recovery works on generated BPSK/QPSK examples.

Remaining: Real recording frame recovery; arbitrary CRC solving, field classification accuracy, schema export and understandable Bit-Lens labels.

Done when: R-10 frame and field targets pass on a held-out frame corpus.

Expected GUI: automated spectral feature analysis [PARTIAL]

Today: Engine-backed waterfall, constellation, proof and partial undecided view; curated generated proof samples.

Remaining: PSD/constellation/waterfall first-view acceptance, clear pipeline affordances, real result labels, export/verify UX, responsive 100 MB handling and accessibility checks.

Done when: A judge reaches each PS clause in ≤2 clicks; R-11/R-12 metrics measured on reference hardware.

3. Functional requirement backlog (R-01 to R-17)

These are the PRD acceptance gates. Where an operator exists only with a supplied recipe, its blind identification gate remains open.

R-01 · File ingest

Complete WAV 8–32 bit integer/float, extensible, RF64/BW64, auxi/Kiwi; SigMF 1.x; raw u8/i8/i16/i32/f16/f32/f64 × endian × IQ order with confidence and override; stream to 8 GB. Gate: 100% correct on 200 benchmark files ≥64 kB at SNR ≥0 dB; clean fuzzing.

R-02 · Sampling frequency

Check declared versus measured Fs; backsolve from standard clock/anti-alias edge when possible. Gate: P95 ≤10 ppm on eligible ≥10 s, ≥10 dB anchored files; otherwise report Fs unknown.

R-03 · Continuous parameters

Finish family-specific symbol-rate estimators and 95% intervals for rate, center, bandwidth, SNR, roll-off, FSK tones/index and burst timing. Gate: rate P95 ≤1e-4 relative, SNR ≤1 dB, bandwidth ±5%, interval coverage 0.95±0.03.

R-04 · Blind modulation

Add OQPSK, 8PSK, 16/64QAM, 2/4/8-FSK and MSK to fused, calibrated open-set identification; keep NOVEL/ambiguous output. Gate: ≥98% top-1 for ≥10 dB and ≥1,000 symbols; measure coverage and calibration error.

R-05 · Blind FEC

Recover convolutional rate/K/generators/puncture, RS n/k/basis/depth, LDPC catalogue and concatenation; resolve phase ambiguity. Gate: ≥95% ID in specified BER/data envelopes, including curves to BER 1e-2.

R-06 · Blind interleaver

Joint soft search over block, convolutional, diagonal and pseudo-random periods; report ambiguity and insufficient data. Gate: ≥95% ID for periods ≤1,024 with ≥$3N^2$ coded bits; ≥90% decision rate; rows needed ≤3N.

R-07 · Demodulation

Integrate FSK; add OQPSK/MSK, burst mode, CMA→DD-LMS equalizer and soft-LLR ambiguity sets. Gate: AWGN BER within 0.5 dB of theory at 1e-4, HF channel curves, burst detection ≥99% at ≥10 dB.

R-08 · Four inverse interleavers

Integrate finite-stream convolutional inverse and manual override into the same recipe/proof path. Gate: 10,000 bit-exact round trips per type.

R-09 · FEC decode

Expand punctured soft Viterbi, RS dual basis, concatenation and LDPC variants; compare independent oracles. Gate: FER 1e-3 point within 0.2 dB of reference, by family.

R-10 · Frame and field discovery

Add soft sync, bit correlation, raster, entropy, counters, length and GF(2) CRC solving for real payloads. Gate: 100% frame length/sync at ≥20 frames, BER ≤1e-3; ≥95% field-class accuracy.

R-11 · GUI visibility

Show waterfall, PSD and constellation immediately; make each PS step reachable in ≤2 clicks; finish visual, readability, responsive and accessibility gates. Gate: 100 MB interactive <2 s plus visual regression.

R-12a · One-click fast path

Bound and record at most eight proof attempts per emitter; connect full chain automatically. Gate: P95 <10 s per million samples in the nominal envelope.

R-12b · Deep path

Add anytime progress, best-so-far result and bounded wider search (periods ≤4,096, LDPC catalogue, scrambler degree ≤23). Gate: default 10 min cap and reproducible interruption.

R-13 · Python/C++/GNU Radio

Keep Python orchestration; build/benchmark required C++ kernels; validate GNU Radio target/oracle and document boundaries. Gate: each technology is actually exercised for its PRD purpose.

R-14 · Portable output

Complete PDF+JSON certificate, SigMF annotations, bits, header/payload CSV, versioned recipe JSON and runnable .grc. Gate: bit-exact export and verifier replay on a second machine.

R-15 · Signal triage

Validate digital/voice/CW/noise classifier and short-file abstention. Gate: ≥95% over 400 independent labelled files in <1 s.

R-16 · Headless/batch

Finish analyze, verify, batch, gen and bench sharing one pipeline with the API. Gate: identical recipes on the benchmark corpus and batch error recovery.

R-17 · Multi-emitter scene

Build OS-CFAR detection, polyphase channelization and burst segmentation; distinguish overlapping emissions. Gate: ≥95% emitters at ≥6 dB detected; SNR loss ≤0.3 dB.

4. USP completion: what can be shown, what still needs proof

C1 · Layered proof and independently verifiable certificate [PARTIAL]

Today: Layer verdicts, signed JSON, source-file recheck, near-miss rejection and gated export run for generated captures.

Remaining: Calibrate thresholds on disjoint data; include real-channel noise references; freeze and run ≥2,000,000 stratified near-miss proof attempts; compare independent decoders on real recordings; reproduce on another machine. Current false_pass_bound is null and calibration_id is v0-defaults.

Done when: A certificate reports only measured bounds, attempt count, calibration ID and honest nondeterminable layers. No numerical false-pass claim before the campaign.

C2 · Data sufficiency and collection tasking [PARTIAL]

Today: Ledger and generated examples can express an undecided state; unsupported real framing no longer produces an invented duration.

Remaining: Calibrate predictive P50/P90 extra-duration distributions across families, durations, SNR and sensors; produce actionable center frequency, bandwidth, sample rate, duration, minimum SNR and sensor recommendation with uncertainty.

Done when: Held-out predictive coverage and duration-error KPI in PRD §3.4 pass; impossible/unsupported cases say so instead of promising a longer recording.

C3 · Evidence stitching [PARTIAL]

Today: Three generated sensors with a decoy can be pooled and bad association rejected.

Remaining: GPS-timestamped ≥3-site real simultaneous HF captures; align on sync marker, normalize clocks/noise, handle gaps and reject incorrect matches; quantify false merges.

Done when: Equal-total-data alignment loss ≤10% and false merge ≤1% on a held-out multi-site set; each capture retains its own proof.

C4 · Sensor Twin [PARTIAL]

Today: Lite clock/noise estimates exist as inputs to proof and stitching.

Remaining: One GPS-disciplined reference to fix clock gauge; weighted network clock solution and uncertainty; measured noise shape and independent sensor transfer. Full IQ imbalance/phase-noise/clipping Twin is later P2 scope.

Done when: Clock/noise uncertainty is calibrated on real sensors; without a reference the UI reports only relative clocks.

C5 · Recipe Compiler [PARTIAL]

Today: Certified generated recipes produce a standalone receiver package, versioned data and a .grc file; NumPy receiver reproduces generated bits.

Remaining: Execute the generated GNU Radio 3.10 graph in a pinned offline compiler pack against the source and a second real sensor; check detector behavior and packaging.

Done when: The graph reproduces certified bits exactly on source; independently replays on another machine. Do not call the graph validated while it has only been generated.

D1 · Soft joint coding engine [PARTIAL]

Today: The demonstrated blind path combines convolutional FEC with block geometry; broader inverse operators need supplied recipes.

Remaining: Soft rank profile, joint interleaver/code search, peeling, ambiguity and post-FEC scrambler cosets, RS basis/depth; retain a measured data-sufficiency exit.

Done when: Sealed noisy cases and a real case recover the same recipe without analyst hints; measure R-05/R-06 decision and data thresholds.

D2 · Blind scrambler recovery [PARTIAL]

Today: Additive scrambler search works on the generated frame family.

Remaining: Recover placement before and after FEC, polynomial/seed by Berlekamp–Massey and bounded seed search; reject equivalent hypotheses honestly. Self-synchronizing types are P2.

Done when: A held-out mixed-placement corpus recovers polynomial, seed and placement; CRC/proof rejects wrong scramblers.

D3 · Bit-Lens and CRC solving [PARTIAL]

Today: The UI presents frame structure and current engine correlates sync, counters and CRC on generated examples.

Remaining: Port any-CRC GF(2) solver to the live engine and UI, distinguish header/payload fields, explain bit-correlation view in plain language, export the inferred schema.

Done when: Field/CRC truth suite meets R-10 and a judge can explain a real frame without reading raw bitmaps.

D4 · Scene [PARTIAL]

Today: Single-emitter analysis is the demonstrated path.

Remaining: OS-CFAR detection, per-emitter channelization and burst segmentation are R-17 P1; successive interference cancellation is P2.

Done when: Two-or-more-emitter recording separates tracks and meets detection/SNR-loss gates without mixing their certificates.

D5 · Signal Genome [PARTIAL]

Today: Some standard labels and anchors inform current output, without a validated sample-rate backsolve catalogue.

Remaining: Build a small versioned, cited standards table for clock/rate/frame hypotheses and use it only as an explicitly labelled prior; the approximately 30-standard table is P2.

Done when: Known standards help resolve Fs/recipe on held-out files, while novel signals still abstain rather than force a standard label.

D6 · HF robustness [PARTIAL]

Today: Public HF files are measured and classified, but their full payloads are not recovered.

Remaining: Watterson fading curves with ground truth, then real HF tuning and independent reference-decoder agreement; rig emulation is P2.

Done when: Report BER/FER and undecided rates by channel/SNR, including at least one full real HF decode if this is claimed.

D7 · Trust layer [PARTIAL]

Today: The UI and certificate expose partial and undetermined states; current thresholds are default, not calibrated.

Remaining: Calibrate per-family probabilities/intervals, split-conformal sets and NOVEL output; preserve uncertainty on every measured value. Energy/Mahalanobis and weighted conformal extensions are P2.

Done when: Held-out coverage/calibration targets pass, and unsupported files never receive a false precision or proof verdict.

C6–C8 · Future crowns [P2/P3]

Today: Emitter Passport, advanced WAV-Lifting and a proof-gated learning flywheel are PRD P2/P3 proposals, not current prototype claims.

Remaining: C6: reconstruct emitter identity and chronology only from supported evidence; C7: lift only physically invertible audio chains; C8: curate training labels only from independently verified proof. Define separate validation and privacy/data gates before implementation.

Done when: These remain clearly labelled future work unless independently built and validated; they do not count toward current PS coverage.

5. Evidence package required for a credible claim

• A frozen, versioned labelled corpus with generator seeds and real recording sites split across train, calibration and test; provenance, SNR, duration, modulation, FEC, interleaver and ground truth recorded.

• One sealed synthetic end-to-end case for every claimed family, plus independent real captures with external decoder agreement. A classifier label alone is not a recovered payload.

• A per-family confusion matrix, abstention rate, calibrated intervals, BER/FER curves, decision rate, latency and peak RAM on the PRD 8-core/16 GB/no-GPU laptop; sample counts and confidence intervals beside each number.

• Near-miss methodology and raw results: stratified wrong recipes, frozen thresholds, at least 2,000,000 attempts before claiming the planned false-pass bound; distinguish zero observed errors from zero risk.

• Second-machine signed-certificate replay with tampered-certificate and tampered-recording rejection; GNU Radio bit-exact replay; reproducible export manifest.

• Judge path: bundled sample → measured signal → decoded bits → explained header/payload → proof layers → exported certificate → verifier; an unsupported upload ends with clear measured evidence and an accurate reason.

6. Product and delivery gates (NFR-01 to NFR-13)

NFR-01–03 · Platform and scale

Clean Windows 10/11 and Ubuntu 22.04/24.04 builds; 8 GB memory-mapped processing; ≤4 GB engine RAM; 8-core/16 GB/no-GPU reference test.

NFR-04–06 · Reliability and trust

Engine ready ≤5 s; UI reconnect/resume; localhost-only API, packet capture confirms zero outbound traffic; deterministic recipe and canonical certificate body.

NFR-07–09 · Install and operations

Per-user offline installers ≤600 MB per OS including offline WebView2 on Windows; cancel/watchdog/process isolation; rotated local logs and support bundle.

NFR-10–13 · Accessibility and supply chain

WCAG AA/reduced-motion audit; licence register with no copyleft shipped engine code; hashed dependency pins/SBOM; offline dependency kill matrix preserving honest verdicts.

7. Order of work and submission decision

Before tomorrow's submission: lock a truthful, repeatable judge demonstration. Re-run the full engine check and UI build; manually traverse every curated sample and the certificate verifier; ensure sample import selects only the recording/sidecar, while certificate choice remains manual. Make all unsupported examples clearly undecided. Put the coverage table and explicit limitations in the pitch. Capture screenshots/video and a short evidence log. These steps improve reliability without implying the unbuilt families are complete.

Next engineering gates: (1) wire FSK and convolutional interleaver through recipe/search/proof/API; (2) obtain a full real-file decode with external decoder agreement; (3) extend blind QAM/PSK and FEC/interleaver families; (4) calibrate uncertainty and near-miss proof; (5) run GNU Radio and multi-site real validation; (6) finish ingestion scale, scene, performance, installers and supply-chain gates. Dependencies matter: do not advertise a broad certified family before blind recovery and its proof have both passed.

> Submission wording: “SANJAYA is a working proof-driven signal-analysis prototype. It demonstrates blind BPSK/QPSK recovery, independent certificate recheck, explicit undecided outcomes, generated multi-sensor stitching and proof-gated receiver export. The remaining PS families and quantitative KPIs are tracked as the next validation gates.”

Source of scope and targets: PRD.md §§2.1, 3–5; source of demonstrated status: PROTOTYPE_AUDIT.md (28 Sep 2026) and the latest 108-test check. Targets in this document are acceptance criteria, not measured results.